

PD envelope

Th-Def (A, I, γ) PD ring, $B: A\text{-alg}$, $J \triangleleft B$, then $\exists B\text{-alg } P_{B, \gamma}(J)$

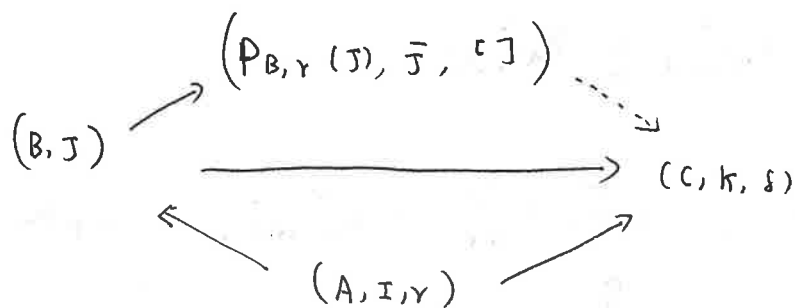
w/ PD ideal $(\bar{J}, [\cdot])$ s.t. $J \subset P_{B, \gamma}(J) \subset \bar{J}$

satisfying the following

- $[\cdot]$ is compatible with γ

universal property: for any (B, J) -algebra (C, κ) w/ a PD str. δ compatible w/ γ ,

$\exists!$ PD morphism $(P_{B, \gamma}(J), \bar{J}, [\cdot]) \rightarrow (C, \kappa, \delta)$ making the diagram commute,



Reformulation $\gamma\text{PD}-(B, J)\text{-Alg}$: Obj. are (C, κ, δ) , $(C, \kappa): (B, J)\text{-algebra}$, δ PD str. comp. w/ γ .

$$\text{Hom}_{\gamma\text{PD}-(B, J)\text{-Alg}}((P_{B, \gamma}(J), \bar{J}, [\cdot]), (C, \kappa, \delta)) \simeq \text{Hom}_{(B, J)\text{-Alg}}((B, J), (C, \kappa)).$$

$$\gamma\text{PD}-(B, J)\text{-Alg} \xrightleftharpoons[\text{forgetful}]{\text{PD envelope}} (B, J)\text{-Alg}$$

$$\text{Also define } P_{B, \gamma}^n(J) = P_{B, \gamma}(J) / \bar{J}^{[n+1]}$$

Rem. The construction of PD envelope commutes w/ flat base change: $B' \text{ flat } B\text{-alg}, J \otimes B' \simeq J \otimes_B B'$

(S, I, γ) PD scheme, X (separated) S -scheme. $\Delta: X \hookrightarrow X \times_S X$

Def. $P_{X/S, (I)} := \text{PD-envelope of } X \text{ in } X \times_S X. P_{X/S, (I)}^n = n^{\text{th}} \text{ order PD nbhd.}$

Multiplication

Augmentation ideal of $\mathcal{P}_{X/S, (0)}$ has PD-str.

$$\mathcal{P}_{X/S, (0)} \otimes_{\mathcal{O}_X} \mathcal{P}_{X/S, (0)} \quad \text{PD-algebra}$$

$$\delta : \mathcal{O}_X \otimes_{\mathcal{O}_S} \mathcal{O}_X \longrightarrow (\mathcal{O}_X \otimes_{\mathcal{O}_S} \mathcal{O}_X) \otimes_{\mathcal{O}_X} (\mathcal{O}_X \otimes_{\mathcal{O}_S} \mathcal{O}_X)$$

$$x \otimes y \longmapsto x \otimes 1 \otimes 1 \otimes y$$

univ. prop

of PD envelope

$$\delta : \mathcal{P}_{X/S, (0)} \longrightarrow \mathcal{P}_{X/S, (0)} \otimes_{\mathcal{O}_X} \mathcal{P}_{X/S, (0)}.$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \mathcal{P}_{X/S, (0)}^{m+n} & \xrightarrow{\delta^{m,n}} & \mathcal{P}_{X/S, (0)}^m \otimes_{\mathcal{O}_X} \mathcal{P}_{X/S, (0)}^n \end{array}$$

$$\mathcal{D}_{X/S, n}^{(0)} := \text{Hom}_{\mathcal{O}_X} (\mathcal{P}_{X/S, (0)}^n, \mathcal{O}_X), \quad \mathcal{D}_{X/S}^{(0)} = \bigcup_{n \geq 0} \mathcal{D}_{X/S, n}^{(0)} \quad \text{algebra}$$

$\uparrow$

sheaf of diff. ops "of level 0"

Digression (EGA IV, Grothendieck) X separated scheme $/S$, $X \xrightarrow{\mathcal{I}_\Delta} X \times_S X$

$$\mathcal{P}_{X/S}^n := \mathcal{O}_{X \times_S X} / \mathcal{I}_\Delta^{n+1} \quad \text{sheaf of principal parts of order } n$$

$$\mathcal{D}_{X/S, n} := \text{Hom}_{\mathcal{O}_X} (\mathcal{P}_{X/S}^n, \mathcal{O}_X), \quad \mathcal{D}_{X/S} = \bigcup_{n \geq 0} \mathcal{D}_{X/S, n}$$

$\uparrow$

sheaf of diff. ops (of level n)

X smooth over $S \hookrightarrow$ local coord. $t^1, \dots, t^d$. $\tau^i = 1 \otimes t^i - t^i \otimes 1$

$\{\tau^{[k]} : \underline{k} \in \mathbb{N}^d, |\underline{k}| \leq n\}$ is a basis as $\mathcal{O}_X$ -module of $\mathcal{P}_{X/S, (0)}^n$

$\{\tau^{\underline{k}} : \underline{k} \in \mathbb{N}^d, |\underline{k}| \leq n\}$ is a basis as $\mathcal{O}_X$ -module of $\mathcal{P}_{X/S}^n$

Dual basis $\{\partial^{\underline{k}} : \underline{k} \in \mathbb{N}^d, |\underline{k}| \leq n\}$ of $D_{X/S, n}^{(0)}$, $\partial^{\underline{k}} \cdot \partial^{\underline{k}'} = \partial^{\underline{k} + \underline{k}'}$

Dual basis $\{\partial^{[\underline{k}]} : \underline{k} \in \mathbb{N}^d, |\underline{k}| \leq n\}$ of $D_{X/S, n}$. $\partial^{[\underline{k}]} \partial^{[\underline{k}']} = \binom{\underline{k} + \underline{k}'}{\underline{k}} \partial^{[\underline{k} + \underline{k}]}$

Def. $\mathcal{E}$ $\mathcal{O}_X$ -module, A "PD stratification on $\mathcal{E}$ " is a collection of isoms

$$\mathcal{E}_n: P_{X/S, (0)}^n \otimes_{\mathcal{O}_X} \mathcal{E} \xrightarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_X} P_{X/S, (0)}^n \quad \text{s.t.}$$

1) Each $\mathcal{E}_n$ is $P_{X/S, (0)}^n$ -linear

2) $\mathcal{E}_n$'s compatible in the obvious sense, $\mathcal{E}_0 = \text{id}_{\mathcal{E}}$

3) Cycle condition

$$P_{X/S, (0)}^n \otimes P_{X/S, (0)}^m \otimes \mathcal{E} \xrightarrow{\mathcal{E}_{n+m}(\mathcal{E}_{m+n})} \mathcal{E} \otimes P_{X/S, (0)}^n \otimes P_{X/S, (0)}^m$$

$$\downarrow \text{id} \otimes \mathcal{E}_m \qquad \qquad \qquad \uparrow \mathcal{E}_n \otimes \text{id}$$

$$P_{X/S, (0)}^n \otimes \mathcal{E} \otimes P_{X/S, (0)}^m$$

Def. An "HPD stratification on $\mathcal{E}$ " is an isom. $\mathcal{E}: P_{X/S, (0)} \otimes \mathcal{E} \xrightarrow{\sim} \mathcal{E} \otimes P_{X/S, (0)}$

s.t. 1) $\mathcal{E}$ is $P_{X/S, (0)}$ -linear

2) $\mathcal{E} \bmod \bar{J} = \text{id}$

3) Cycle condition.

The def. of

Th. A PD strat. on $\mathcal{E}$ is the same as a left $D_{X/S}^{(0)}$ -mod. str. on $\mathcal{E}$

extending the $\mathcal{O}_X$ -mod str. Also equiv. to that of an integrable connection ∇ on $\mathcal{E}$.

Grothendieck: A "crystal" is char. by the properties:

it is rigid, and it grows.

Any sheaf F on $\text{Cris}(X/S)$ "grows" over PD thickening (U, T, δ) of $U \subset X$ by def'n.

Def. A crystal of $\mathcal{O}_{X/S}$ -mod is a sheaf F of $\mathcal{O}_{X/S}$ -modules s.t.

for any morphism $u: (U', T', \delta') \rightarrow (U, T, \delta)$ in $\text{Cris}(X/S)$, the map

$$u^* F_{(U, T, \delta)} \xrightarrow{\sim} F_{(U', T', \delta')} \text{ is an isom.}$$

Ex. $\mathcal{O}_{X/S}$ is a crystal.

γ extends, as always

Prop. $i: Y \hookrightarrow X$ closed immersion of (S, I, γ) -schemes, then



$i_{\text{Cris}*}$ is exact, and $i_{\text{Cris}*}(\mathcal{O}_{Y/S})$ is a crystal of $\mathcal{O}_{X/S}$ -alg.

Pf. $T = (U, T, \delta) \in \text{Cris}(X/S) \rightsquigarrow \underline{T} \in (X/S)_{\text{Cris}}$.

Claim. $i^*(\underline{T}) \in (Y/S)_{\text{Cris}}$ is representable.

$U_0 = U \cap Y \hookrightarrow T$. Suppose $U \hookrightarrow T$ is defined by PD ideal (J, δ) ,

δ extends to a PD str. $\bar{\delta}$ on $J_1 = I\mathcal{O}_T + J$.

Lemma $i^*(\underline{T})$ is rept'd by $P = (U_0, \mathcal{P}_{U_0, \bar{\delta}}(T), [\gamma]) \in \text{Cris}(Y/S)$.

For $F \in (Y/S)_{\text{Cris}}$, the Zariski sheaf assoc. to $i_{\text{Cris}*}(F)$ on T is

$$i_{\text{Cris}*}(F)_T = \left(\lambda: P \rightarrow T \right)_* (F_P).$$

Pf. P is a PD thickening of U_0 : In general $B/J \not\cong \mathcal{P}_{B, \gamma}(J)/\bar{J}$, but it

is the case when γ extends to B/J : by univ. property, get a section

$$\begin{array}{ccc} \mathcal{P}_{B, \gamma}(J) & \longrightarrow & B/J \\ & \nearrow & \\ & \mathcal{P}_{B, \gamma}(J)/\bar{J} & \end{array}$$

Need to know that $\bar{s}$ extends to $\mathcal{O}_{U_0}$.

image of $J_1 = I\mathcal{O}_T + J$ in $\mathcal{O}_{U_0}$ is just $I\mathcal{O}_{U_0}$, to which γ is assumed to extend. $\checkmark \Rightarrow P \in \text{Cris}(Y/S)$.

P rep. $i^*(T)$: univ. prop. of PD envelope.

$$i_{\text{cris}*}(F)(T) = \text{Hom}_{\text{Cris}(Y/S)}(i^*(T), F) = F(P) \quad \square \text{ Lemma.}$$

exactness of $i_{\text{cris}*}$: underlying Zariski space of $\lambda: P \rightarrow T$

is the closed immersion $U_0 \rightarrow U$, so λ^* is exact.

$i_{\text{cris}*}(\mathcal{O}_{Y/S})$ is a crystal $u: (U', T', S') \rightarrow (U, T, S) \in \text{Mor}(\text{Cris}(X/S))$,

$$u^*(i_{\text{cris}*}(\mathcal{O}_{Y/S})_T) \xrightarrow{\sim} i_{\text{cris}*}(\mathcal{O}_{Y/S})_{T'}$$

$$\Leftrightarrow \mathcal{O}_{T'} \otimes \mathcal{P}_{U \cap Y, \bar{s}}(T) \xrightarrow{\sim} \mathcal{P}_{U' \cap Y, \bar{s}'}(T')$$

Locally, assume $U = U'$, all schemes are affine: $T = \text{Spec } B$, $T' = \text{Spec } B'$.

$u: (B, J, S) \rightarrow (B', J', S')$ PD mor. $U = U' \Rightarrow B/J \xrightarrow{\sim} B'/J'$.

Let $L \triangleleft B$, $L' \triangleleft B'$ define the closed subscheme $U_0 = U \cap Y$

$$\begin{array}{ccc} \mathcal{P}_{B, \bar{s}}(L) \otimes_B B' & \xrightarrow{p} & \mathcal{P}_{B', \bar{s}'}(L') \\ & \nwarrow \quad \nearrow & \\ & B' & \end{array}$$

Claim. p is an isom.

$$\text{Hom}_{\bar{\delta} \text{ PD} - (B', L') - \text{Alg}} \left((P_{B', \bar{\delta}}(L'), \bar{L}', \bar{C}) , (C, \kappa, \delta_C) \right)$$

$$= \text{Hom}_{(B', L') - \text{Alg}} \left((B', L') , (C, \kappa) \right)$$

$$B'/L' \simeq B/L$$

$$\equiv \text{Hom}_{(B, L) - \text{Alg}} \left((B, L) , (C, \kappa) \right)$$

$$B \rightarrow B'$$

$$\cup \quad \cup$$

$$IB+J \rightarrow IB'+J'$$

$$\equiv \text{Hom}_{\bar{\delta} \text{ PD} - (B, L) - \text{Alg}} \left((P_{B, \bar{\delta}}(L), \bar{L}, \bar{C}) , (C, \kappa, \delta_C) \right)$$

WANT

$$\equiv \text{Hom}_{\bar{\delta} \text{ PD} - (B', L') - \text{Alg}} \left(\underbrace{(P_{B, \bar{\delta}}(L), \bar{L}, \bar{C}) \otimes_{(B, IB+J, \bar{\delta})} (B', IB'+J', \bar{\delta}')}_{(B', IB'+J', \bar{\delta}')}, (C, \kappa, \delta_C) \right)$$

need to show this exist.

Exist when $B' \rightarrow B'/IB'+J' \int^\rightarrow$ has a section, i.e. $IB'+J'$ augmented.

this is hard in general.

$P_{Y, r}(x)$, PD envelope

$$\text{Cor. } P_Y(x) \otimes_{\mathcal{O}_X} P_{X/S, (0)} \xrightarrow{\sim} P_Y(X \times_S X), \quad P_{X/S, (0)} \otimes_{\mathcal{O}_X} P_Y(x) \xrightarrow{\sim} P_Y(X \times_S X)$$

$$\text{Pf. } \begin{array}{ccc} X & \longrightarrow & X \times_S X \\ \uparrow i & & \downarrow p_2 \\ Y & \longrightarrow & X \end{array} \quad \begin{array}{ccc} & & P_{X/S, (0)} \\ & \nearrow \pi_1 \downarrow \pi_2 & \\ X & \longrightarrow & X \end{array} \quad \pi_i \in \text{Mor}(\text{cris}(X/S))$$

$$\text{cris}_*(\mathcal{O}_{Y/S}) \text{ is a crystal} \Rightarrow \pi_i^* \text{cris}_*(\mathcal{O}_{Y/S})_X \xrightarrow{\sim} \text{cris}_*(\mathcal{O}_{Y/S}) P_{X/S, (0)}$$

$$\text{cris}_*(\mathcal{O}_{Y/S})_X \simeq P_{Y, r}(x), \quad \text{cris}_*(\mathcal{O}_{Y/S}) P_{X/S, (0)} = P_{Y, \bar{C}}(P_{X/S, (0)})$$

Suffices to show that $P_{Y, \bar{C}}(P_{X/S, (0)}) \simeq P_{Y, r}(X \times_S X)$, Follows from univ. prop.

We have a nat'l isom.

$$\varepsilon: P_{X/S, (0)} \otimes_{\mathcal{O}_X} P_Y(X) \xrightarrow{\sim} P_Y(X) \otimes_{\mathcal{O}_X} P_{X/S, (0)}$$

In fact, ε is an HPD strat. on $P_Y(X)$.

Let $\nabla: P_Y(X) \rightarrow P_Y(X) \otimes_{\mathcal{O}_X} \Omega_{X/S}^1$ be the assoc. int. connection.

$$\nabla(y^{[k]}) = y^{[k-1]} \otimes dy \quad \text{for } y \in V(Y \hookrightarrow X)$$



Th. If $Y \hookrightarrow X$ is a closed imm. of S -schemes w/ X/S smooth,

TFCAE:

(i) the cat. of crystals of $\mathcal{O}_{Y/S}$ -mods on $\text{Cris}(Y/S)$

(ii) the cat. of $P_{Y,r}(X)$ -mod w/ an HPD strat. (as an $\mathcal{O}_X$ -mod)

which is compatible with the canonical HPD strat. of $P_{Y,r}(X)$.

(iii) the cat. of $P_{Y,r}(X)$ -mod w/ an integrable quasi-nilpotent connection

which is compatible w/ the canonical conn. on $P_{Y,r}(X)$. (as an $\mathcal{O}_X$ -mod)

(ii) $\Leftrightarrow$ (iii): Daniel's last talk, also (iii) $\Rightarrow$ (i)

(i) $\Rightarrow$ (ii): Σ is a crystal on $\text{Cris}(Y/S)$

$$p_1, p_2: P_{Y,r}(X_S^\times X) \rightrightarrows P_{Y,r}(X) \in \text{Mon}(\text{Cris}(Y/S))$$

$$\Sigma \text{ crystal} \Rightarrow p_i^*(\Sigma_{P_{Y,r}(X)}) \xrightarrow{\sim} \Sigma_{P_{Y,r}(X_S^\times X)}$$

$$\Rightarrow P_{Y,r}(X_S^\times X) \otimes_{P_{Y,r}(X)} \Sigma_{P_{Y,r}(X)} \simeq \Sigma_{P_{Y,r}(X)} \otimes_{P_{Y,r}(X)} P_{Y,r}(X_S^\times X)$$

$$\text{But } p_{Y,r}(x \times x) \simeq p_Y(x) \otimes_{\mathcal{O}_X} p_{X/S,(0)} \simeq p_{X/S,(0)} \otimes_{\mathcal{O}_X} p_Y(x)$$

$$\sim p_{X/S,(0)} \otimes_{\mathcal{O}_X} \{p_{Y,r}(x)\} \xrightarrow{\sim} \{p_{Y,r}(x)\} \otimes_{\mathcal{O}_X} p_{X/S,(0)} \quad \text{HPD stratification.}$$